

EXPERIMENTS

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COURSE CODE: **DSA0602 DATA HANDLING AND VISUALIZATION**

SET 9 – ONLINE LEARNING ACTIVITY ANALYSIS

Student_ID	Gender	Age	Course	Study_Time (hrs)	Videos_Watched	Quiz_Score	Login_Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.

AIM

To analyze the distribution of quiz scores using a histogram and compare quiz scores across different courses using a boxplot.

PROCEDURE

1. Create or import the Online Learning Activity dataset.
2. Store the student information in R.
3. Plot a histogram for Quiz_Score.
4. Plot a boxplot of Quiz_Score by Course.
5. Add suitable titles and labels.
6. Interpret the student performance.

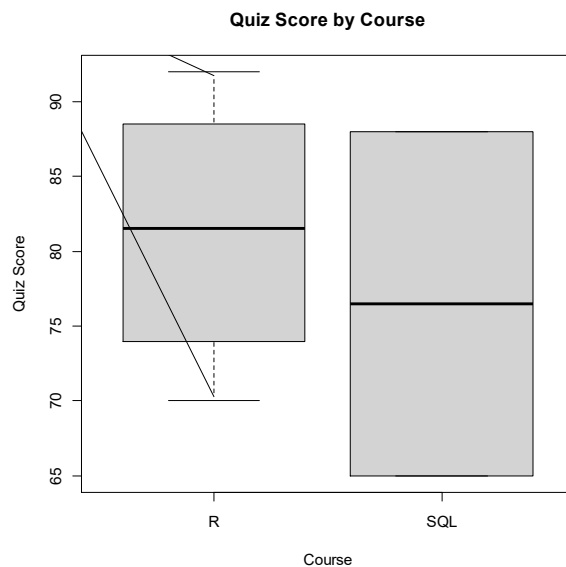
CODE

```
data <- data.frame(
  Student_ID = c("L01","L02","L03","L04","L05","L06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Course = c("R","R","SQL","R","R","SQL"),
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),
  Videos_Watched = c(12,15,8,18,9,14),
  Quiz_Score = c(78,85,65,92,70,88),
  Login_Date = c("2025-01-05","2025-01-05","2025-02-08",
    "2025-02-08","2025-03-12","2025-03-12")
)
```

```
hist(
  data$Quiz_Score,
  main = "Distribution of Quiz Scores",
  xlab = "Quiz Score",
  ylab = "Frequency"
)
```

```
boxplot(
  Quiz_Score ~ Course,
  data = data,
  main = "Quiz Score by Course",
  xlab = "Course",
  ylab = "Quiz Score"
)
```

OUTPUT



RESULT

Thus, the histogram and boxplot were successfully created and student quiz performance was analyzed.

QUESTION 2

Create a scatter plot using R programming between Study_Time and Quiz_Score. Represent Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.

AIM

To analyze the relationship between study time and quiz score using a bubble scatter plot.

PROCEDURE

1. Create the online learning dataset.
2. Use Study_Time on the X-axis.
3. Use Quiz_Score on the Y-axis.
4. Represent Videos_Watched using bubble size.
5. Apply transparency to the bubbles.
6. Add title and axis labels.
7. Interpret the learning pattern

CODE

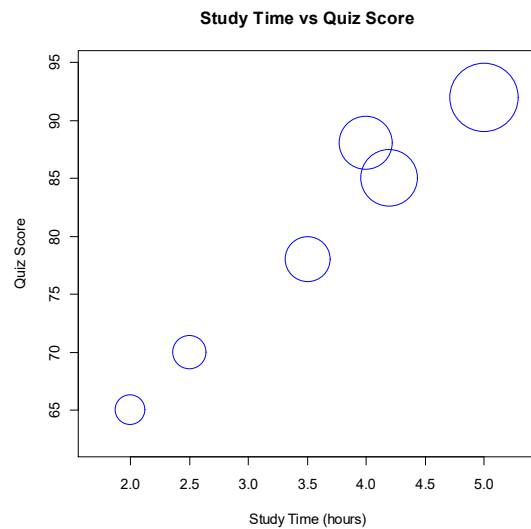
```
data <- data.frame(  
  Student_ID = c("L01","L02","L03","L04","L05","L06"),  
  Gender = c("Male","Female","Male","Female","Male","Female"),  
  Age = c(20,22,19,21,23,20),  
  Course = c("R","R","SQL","R","R","SQL"),  
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),  
  Videos_Watched = c(12,15,8,18,9,14),  
  Quiz_Score = c(78,85,65,92,70,88)  
)  
  
symbols(  
  data$Study_Time,  
  data$Quiz_Score,  
  circles = data$Videos_Watched,  
  inches = 0.4,  
  bg = adjustcolor("blue", alpha.f = 0.4),  
  fg = "blue",
```

```

xlab = "Study Time (hours)",
ylab = "Quiz Score",
main = "Study Time vs Quiz Score"
)

```

OUTPUT



RESULT

Thus, the relationship between study time, videos watched and quiz score was successfully visualized using a bubble scatter plot.

QUESTION 3

Using R programming, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.

AIM

To analyze the monthly average quiz score and identify the performance trend using a moving average line.

PROCEDURE

1. Create the online learning dataset.
2. Convert Login_Date into Date format using as.Date().
3. Extract the month from the login date.
4. Calculate the average quiz score for each month.
5. Calculate a moving average.
6. Plot the monthly average score as a line chart.
7. Add the moving average line.

8. Identify the trend.

CODE

```
data <- data.frame(
  Student_ID = c("L01","L02","L03","L04","L05","L06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Course = c("R","R","SQL","R","R","SQL"),
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),
  Videos_Watched = c(12,15,8,18,9,14),
  Quiz_Score = c(78,85,65,92,70,88),
  Login_Date = c("2025-01-05","2025-01-05","2025-02-08",
    "2025-02-08","2025-03-12","2025-03-12")
)

data$Login_Date <- as.Date(data$Login_Date)

data$Month <- format(data$Login_Date, "%Y-%m")

monthly_avg <- aggregate(
  Quiz_Score ~ Month,
  data = data,
  FUN = mean
)

monthly_avg$Moving_Average <- stats::filter(
  monthly_avg$Quiz_Score,
  rep(1/2, 2),
  sides = 1
)

plot(
  1:nrow(monthly_avg),
  monthly_avg$Quiz_Score,
  type = "o",
  xlab = "Month",
  ylab = "Average Quiz Score",
  main = "Monthly Average Quiz Score",
  xaxt = "n"
)

axis(
  1,
  at = 1:nrow(monthly_avg),
  labels = monthly_avg$Month
)

lines(
  1:nrow(monthly_avg),
```

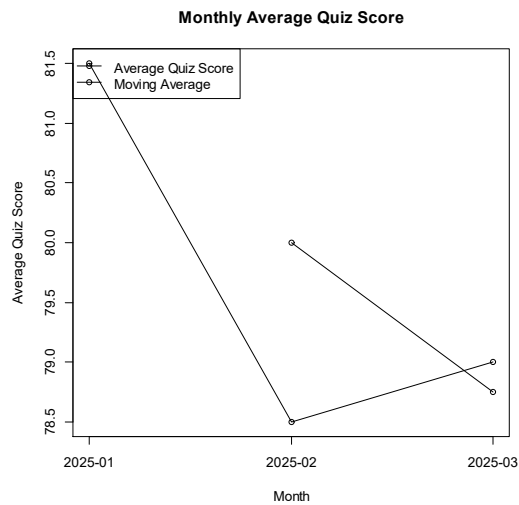
```

monthly_avg$Moving_Average,
type = "o"
)

legend(
  "topleft",
  legend = c("Average Quiz Score", "Moving Average"),
  lty = 1,
  pch = 1
)

```

OUTPUT



RESULT

Thus, the login dates were converted into Date format, monthly average quiz scores were calculated, and the trend was successfully visualized using a line chart with moving average smoothing.

QUESTION 4

Import Online_Learning_Activity.csv into Tableau. Create a bar chart showing average quiz score by Course. Create a scatter plot (Study Time vs Quiz Score). Add filter for Gender and Course. Design a mini interactive dashboard.

AIM

AIM

To create an interactive online learning activity dashboard using R programming with a bar chart, scatter plot, and filters for Gender and Course.

PROCEDURE

1. Create the Online Learning Activity dataset.
2. Load the shiny package.
3. Create a bar chart for average quiz score by course.
4. Create a scatter plot between Study Time and Quiz Score.
5. Represent Videos Watched using bubble size.
6. Add Gender and Course filters.
7. Display the charts and student data in an interactive dashboard.
8. Run the Shiny application.

CODE

```
library(shiny)

data <- data.frame(
  Student_ID = c("L01","L02","L03","L04","L05","L06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Course = c("R","R","SQL","R","R","SQL"),
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),
  Videos_Watched = c(12,15,8,18,9,14),
  Quiz_Score = c(78,85,65,92,70,88),
  Login_Date = c("2025-01-05","2025-01-05","2025-02-08",
    "2025-02-08","2025-03-12","2025-03-12")
)

ui <- fluidPage(
  titlePanel("Online Learning Activity Dashboard"),

  sidebarLayout(
    sidebarPanel(
      selectInput(
        "gender",
        "Select Gender:",
        choices = c("All", "Male", "Female")
      ),

      selectInput(
        "course",
        "Select Course:",
        choices = c("All", "R", "SQL")
      )
    )
  )
)
```

```

    )
  ),

  mainPanel(
    plotOutput("barPlot"),
    plotOutput("scatterPlot"),
    tableOutput("dataTable")
  )
)
)

server <- function(input, output) {

  filteredData <- reactive({
    result <- data

    if (input$gender != "All") {
      result <- result[result$Gender == input$gender, ]
    }

    if (input$course != "All") {
      result <- result[result$Course == input$course, ]
    }

    result
  })

  output$barPlot <- renderPlot({

    result <- filteredData()

    avg_score <- aggregate(
      Quiz_Score ~ Course,
      data = result,
      FUN = mean
    )

    barplot(
      avg_score$Quiz_Score,
      names.arg = avg_score$Course,
      xlab = "Course",
      ylab = "Average Quiz Score",
      main = "Average Quiz Score by Course"
    )
  })

  output$scatterPlot <- renderPlot({

    result <- filteredData()

```



```

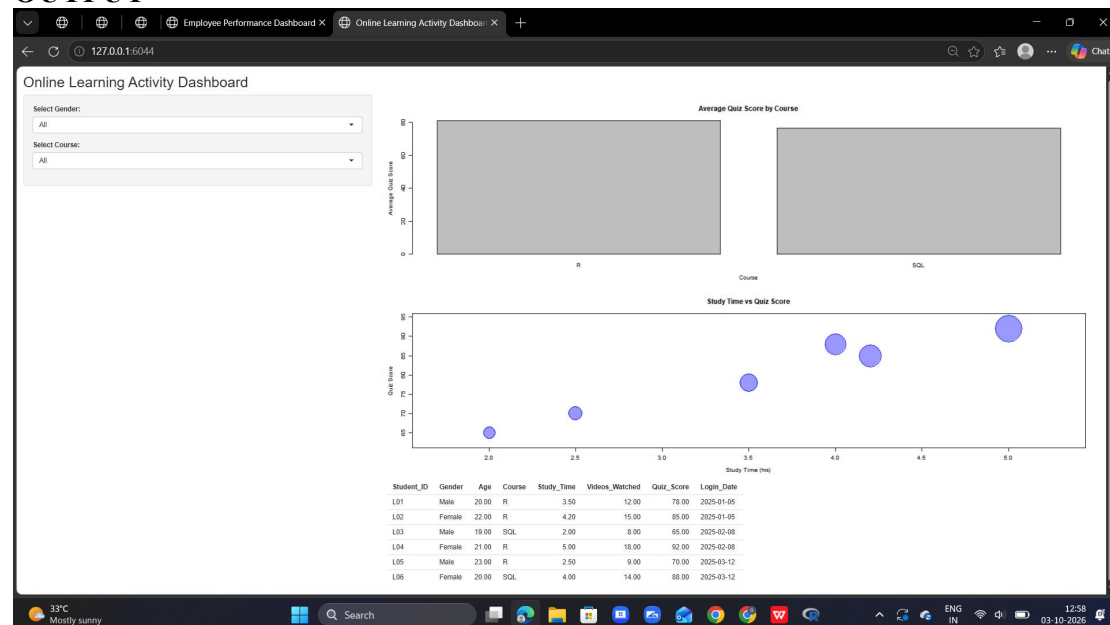
symbols(
  result$Study_Time,
  result$Quiz_Score,
  circles = result$Videos_Watched,
  inches = 0.4,
  bg = adjustcolor("blue", alpha.f = 0.4),
  fg = "blue",
  xlab = "Study Time (hrs)",
  ylab = "Quiz Score",
  main = "Study Time vs Quiz Score"
)
})

output$dataTable <- renderTable({
  filteredData()
})
}

shinyApp(ui = ui, server = server)

```

OUTPUT



RESULT

Thus, an interactive **Online Learning Activity Dashboard** containing the average quiz score bar chart, study-time scatter plot, filters and data table was successfully created using **R Shiny**.